

基础化学 (I)

Fundamentals in Chemistry (I)

苏纪豪

Oct 26rd 2023



常见H的化学位移 (δ)

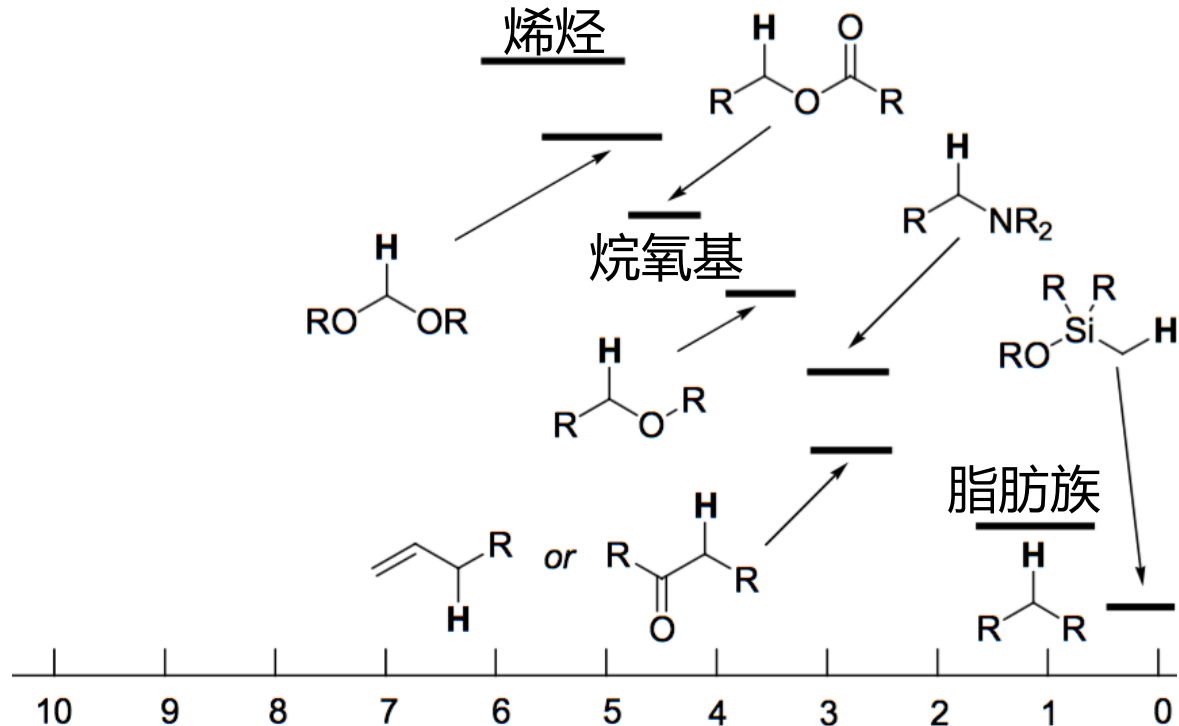
醛氢

芳香化合物

烯烃

烷氧基

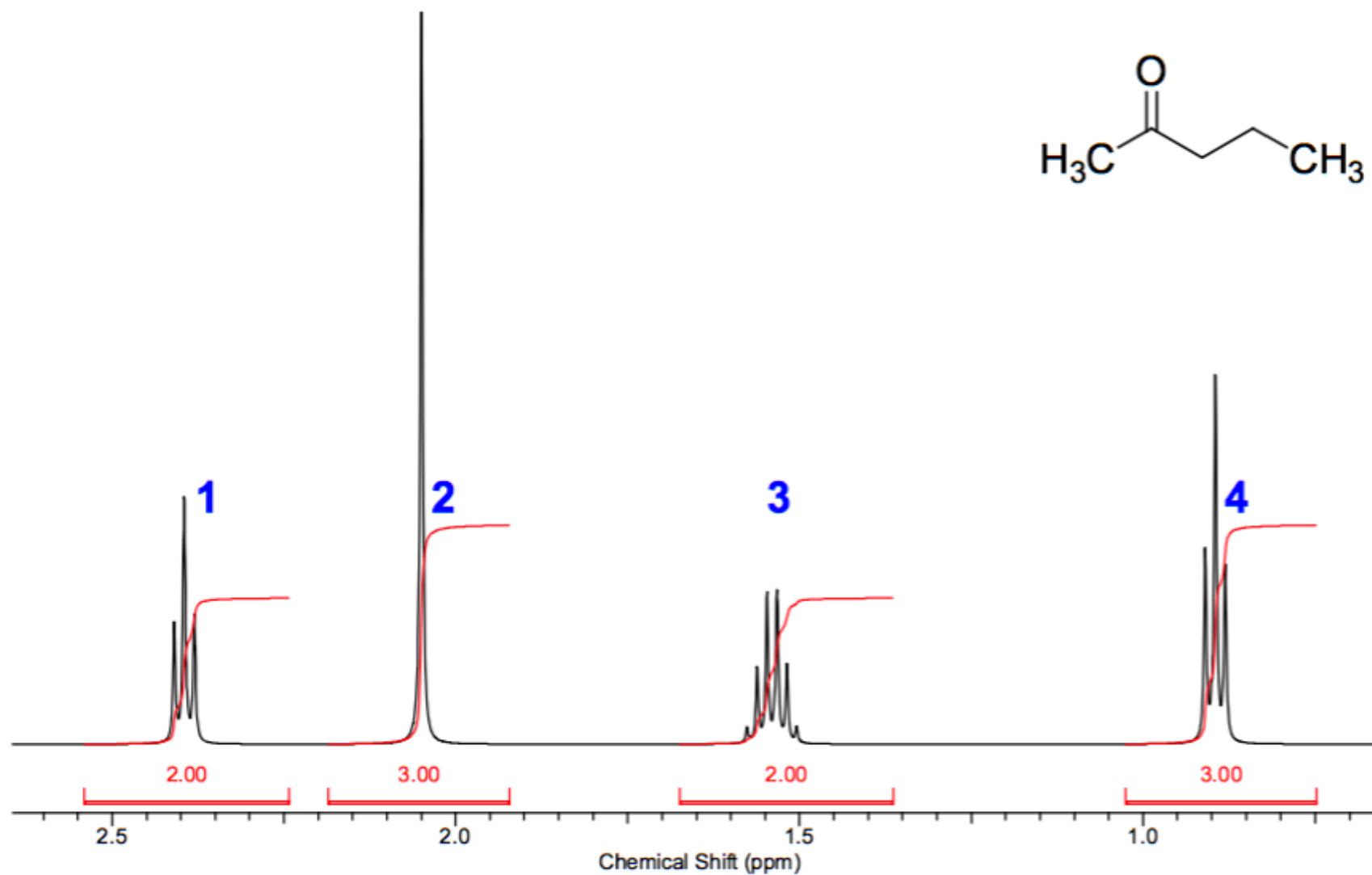
脂肪族



原子核共振的频率取决于真实感受到的局域磁场的实际强度，相同的化合物在不同强度磁场下共振频率不同。通常用内标化合物（TMS）定义化学位移。

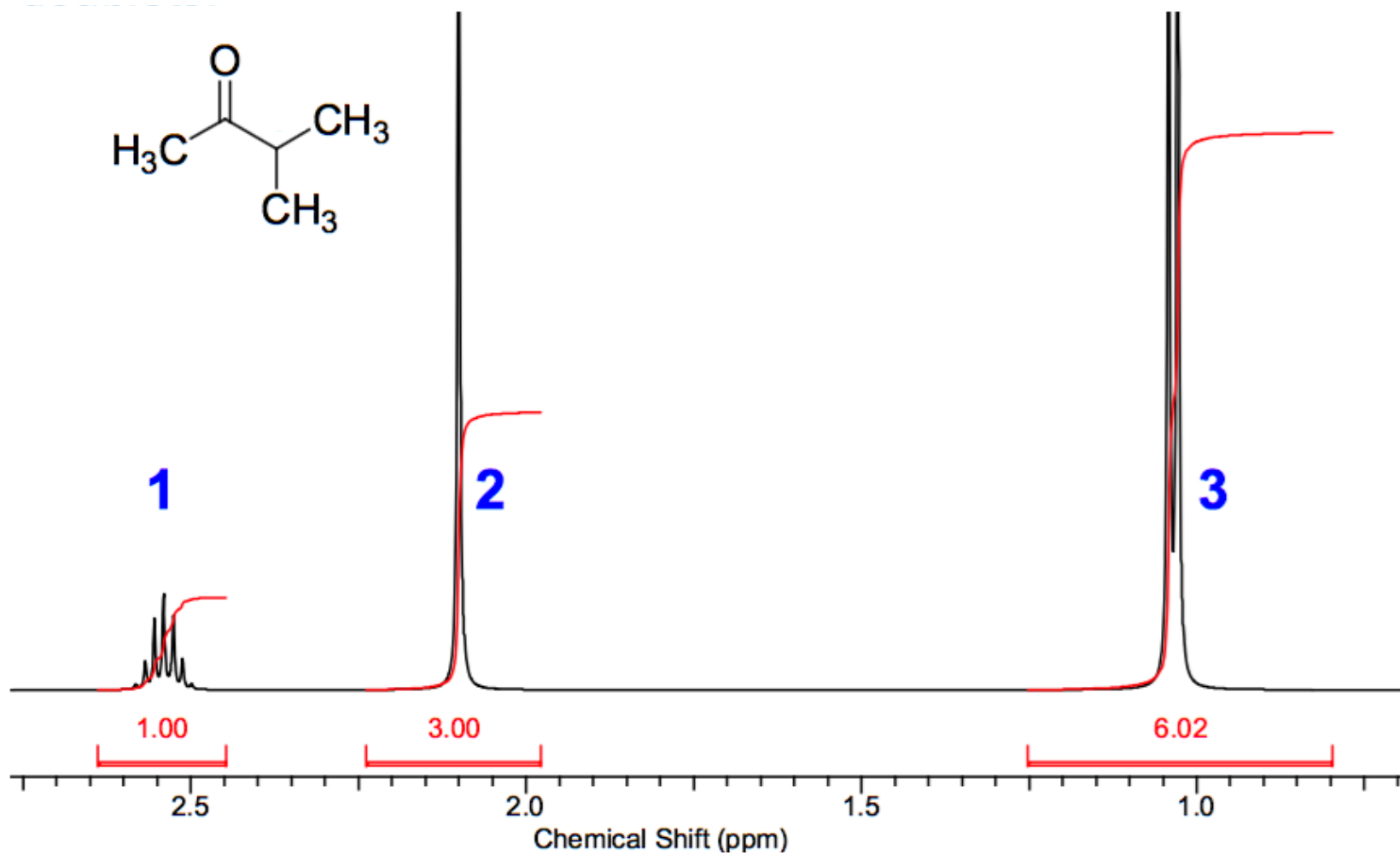


^1H NMR氢谱归属练习



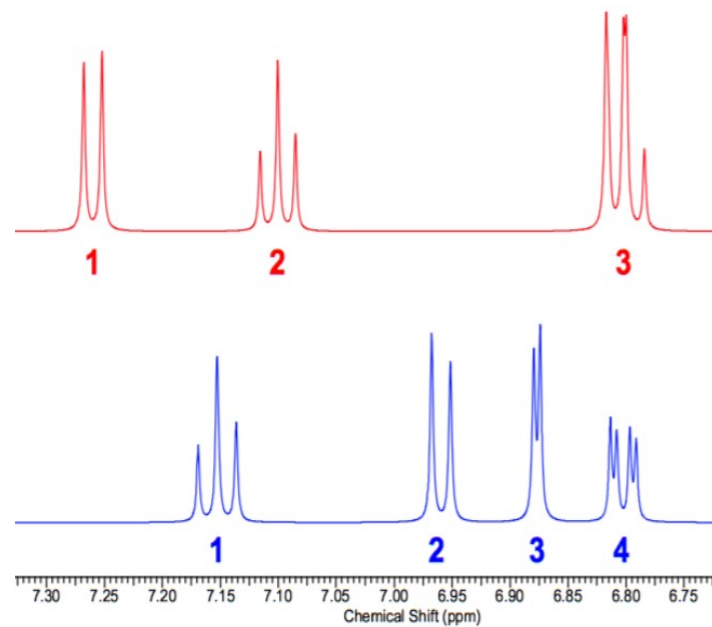
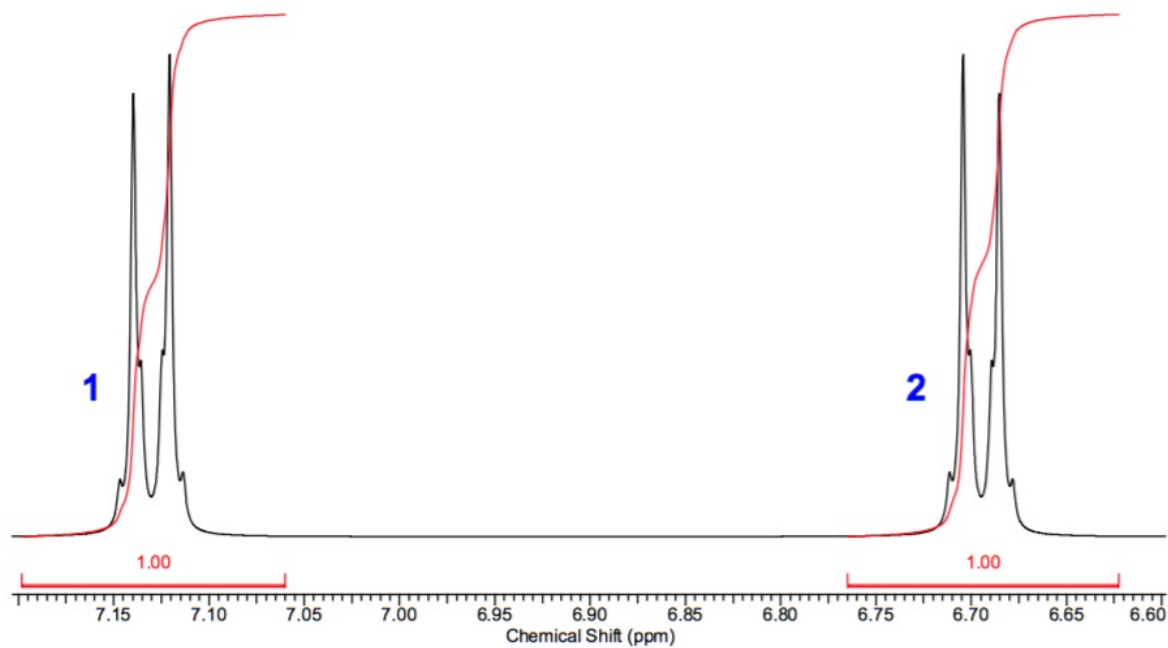
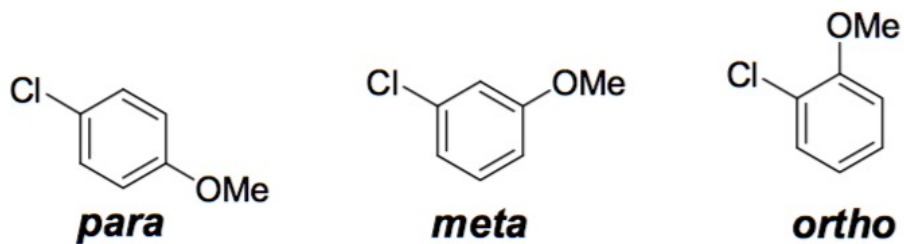


^1H NMR氢谱归属练习



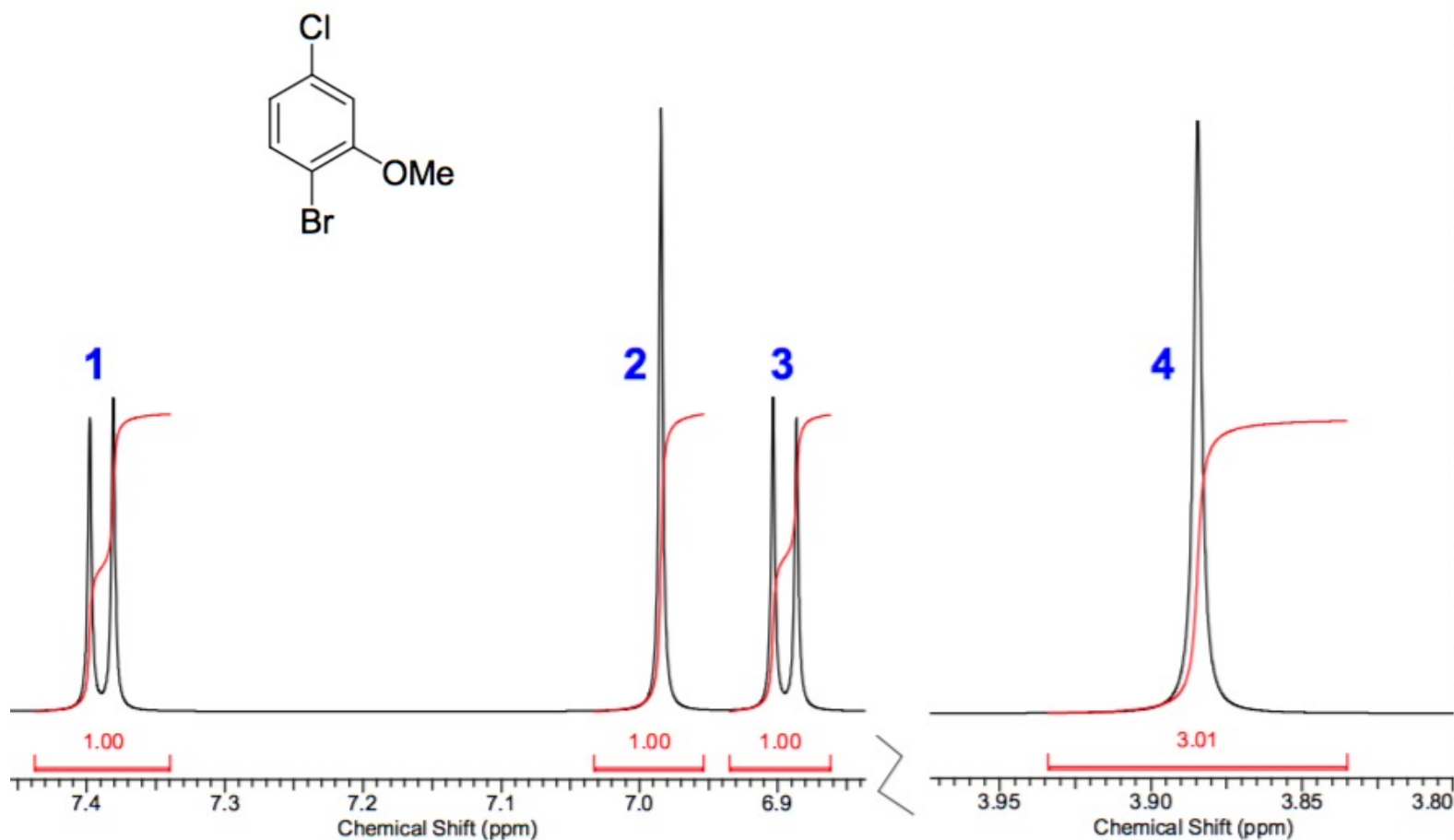


通过 ^1H -NMR区分邻、间、对甲氧基氯苯





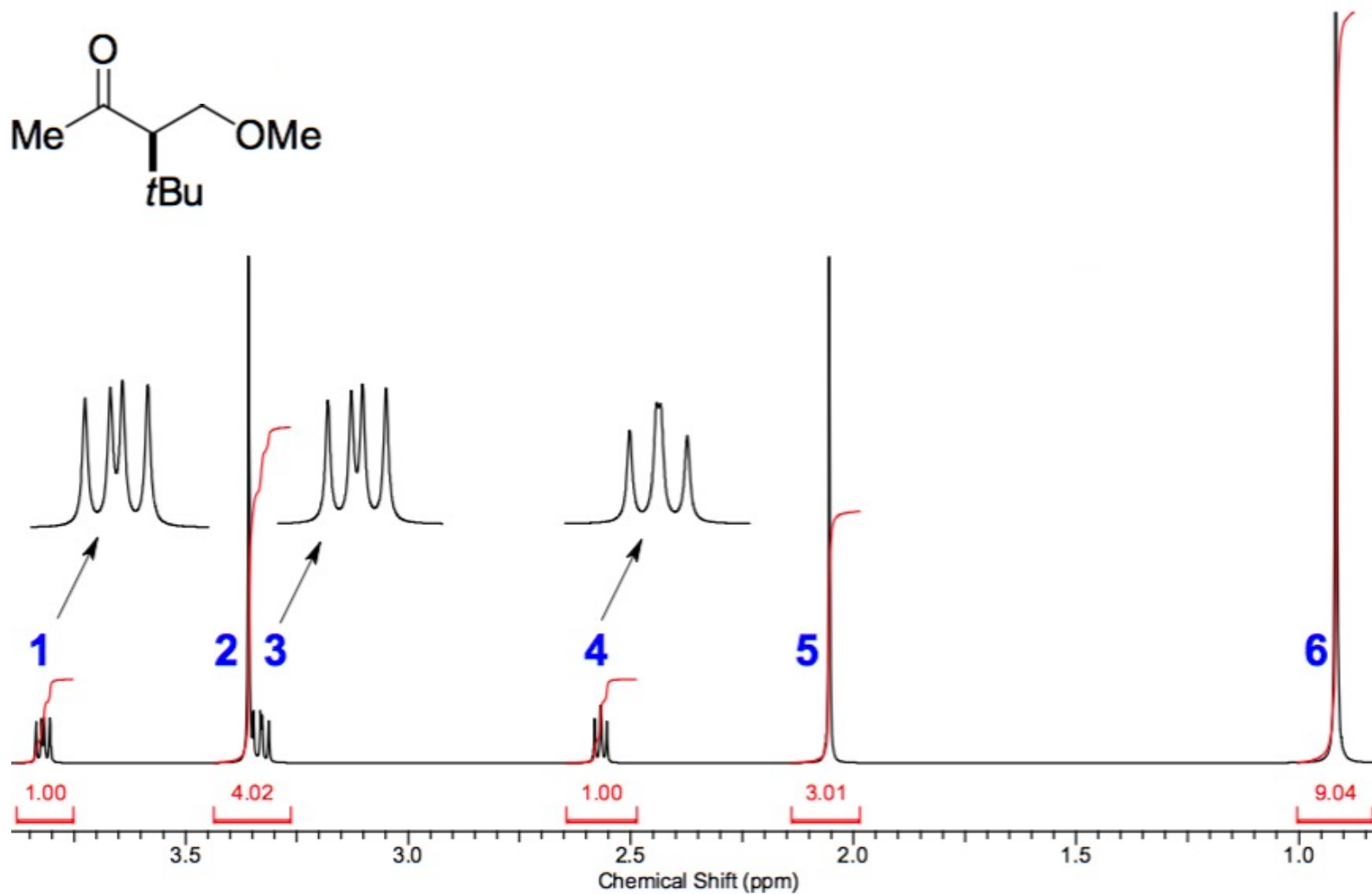
通过 ^1H -NMR确定三取代苯取代基位置



在芳香区，有三个质子。其中两个是双峰，其中一个是单峰。
这意味着是 1,2,5- 取代。



^1H NMR氢谱归属练习





Chemical Equivalence – Which Protons Are “the Same”?

Two protons will not split each other if they have they are "the same"--
i.e., have the same chemical shift.

a pair of protons are:

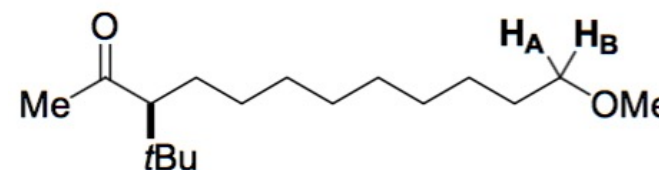
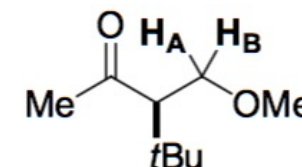
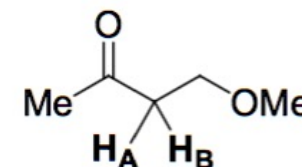
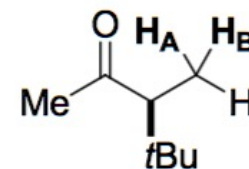
homotopic if they are related by rotation symmetry and are certain to have the same chemical shift ("isochronous") in any medium (test: replace one proton with a deuterium, and find that it does not produce a chiral center);

enantiotopic if they are related by reflection symmetry and will be isochronous in achiral or racemic media (test: replace one proton with a deuterium and find that it produces one of a pair of enantiomers);

diastereotopic if they are neither homotopic or enantiotopic and *could* be anisochronous in achiral or racemic media.

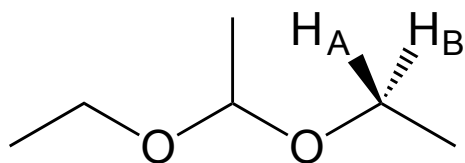
Note that (in achiral or racemic media):

enantiomers have the same NMR spectrum, but
diastereomers have (potentially) different NMR spectra.

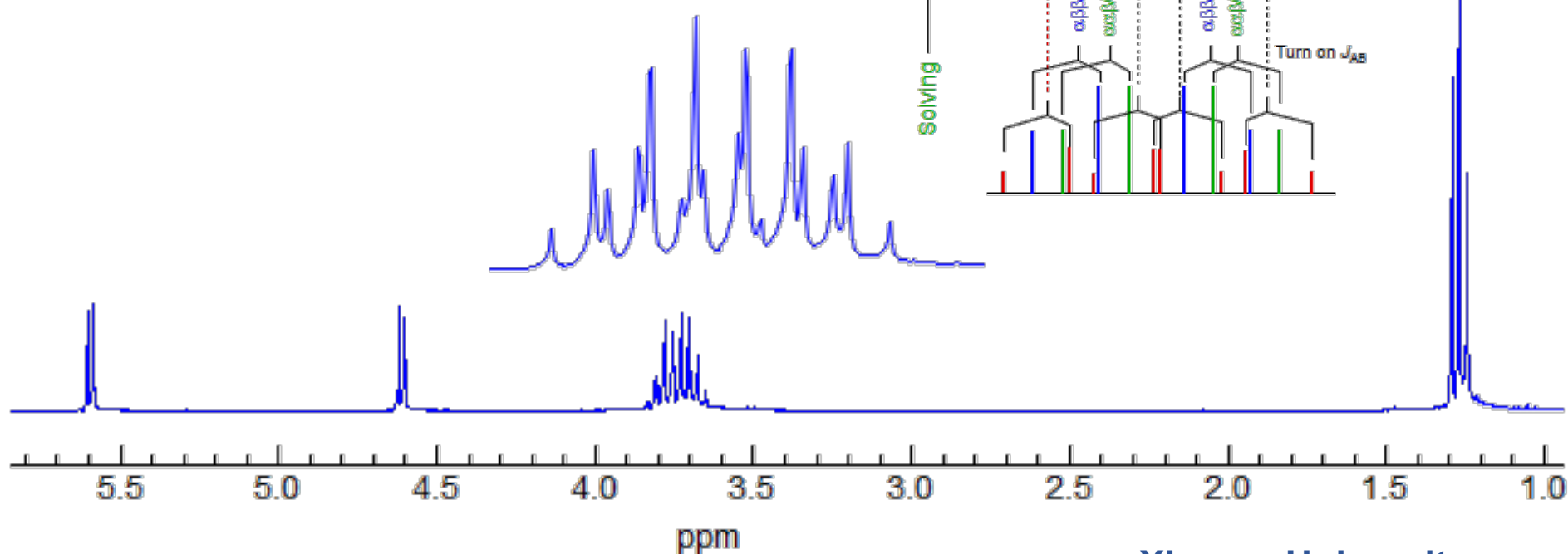
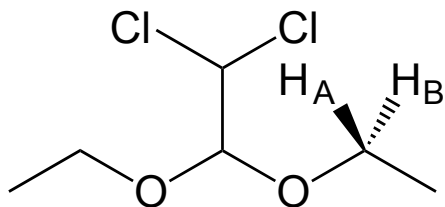




Dichloroacetaldehyde Diethyl Acetal



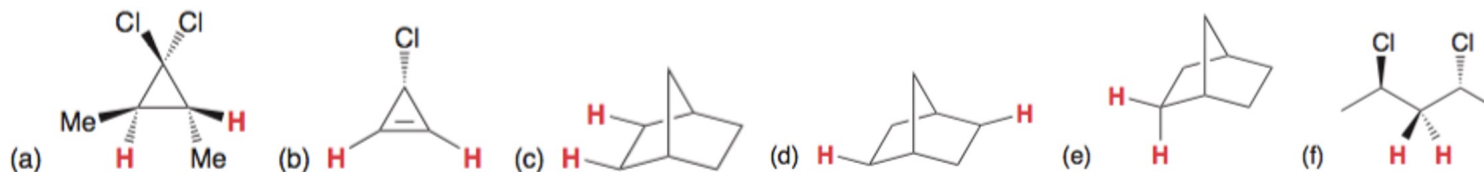
Homotopic? Enantiomeric? Diastereotopic?



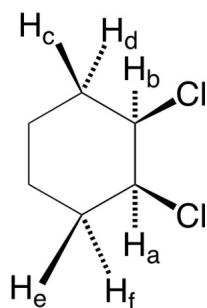


Topicity Practice

Homotopic? Enantiotopic? Diastereotopic?
and which molecules are Chiral?



(1)



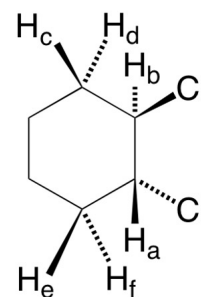
Ha and Hb: _____

Hc and Hd: _____

Hc and He: _____

Hc and Hf: _____

(2)



Ha and Hb: _____

Hc and Hd: _____

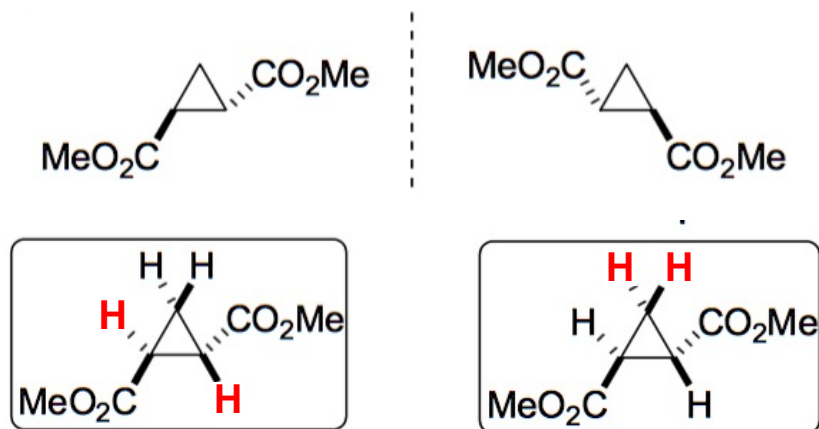
Hc and He: _____

Hc and Hf: _____



Cyclopropane Stereoisomers

The *trans*-isomers are a pair of enantiomers with C_2 -symmetry



homotopic

The *cis*-isomer is *meso*



enantiotopic

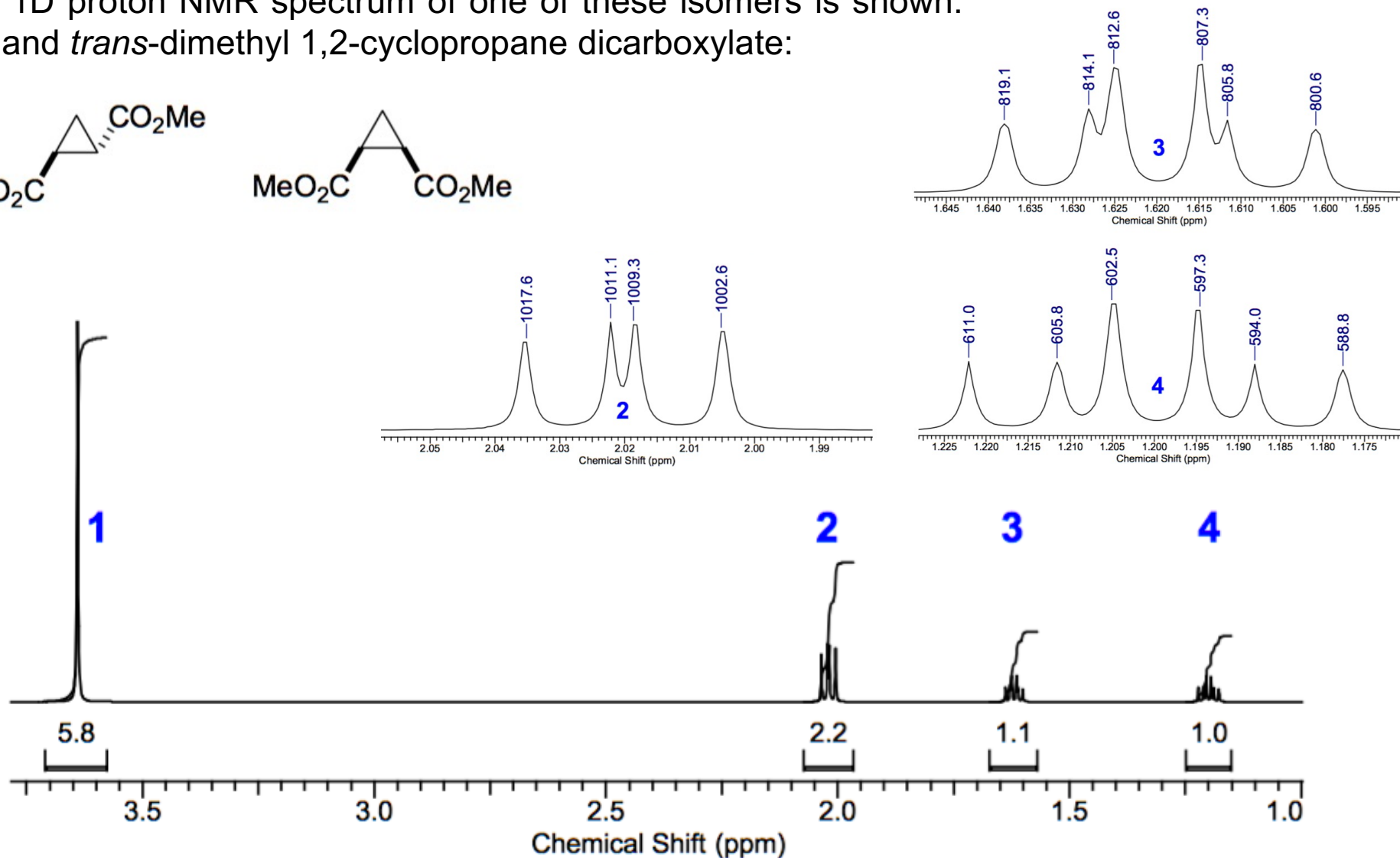
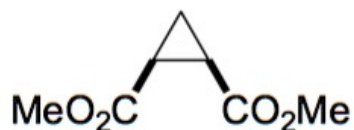
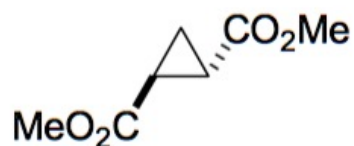
diastereotopic

How many distinct signals should the *cis* isomer have?
What about the *trans*?



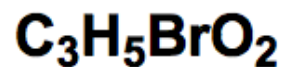
Cyclopropane Stereoisomers

The 1D proton NMR spectrum of one of these isomers is shown:
cis- and *trans*-dimethyl 1,2-cyclopropane dicarboxylate:



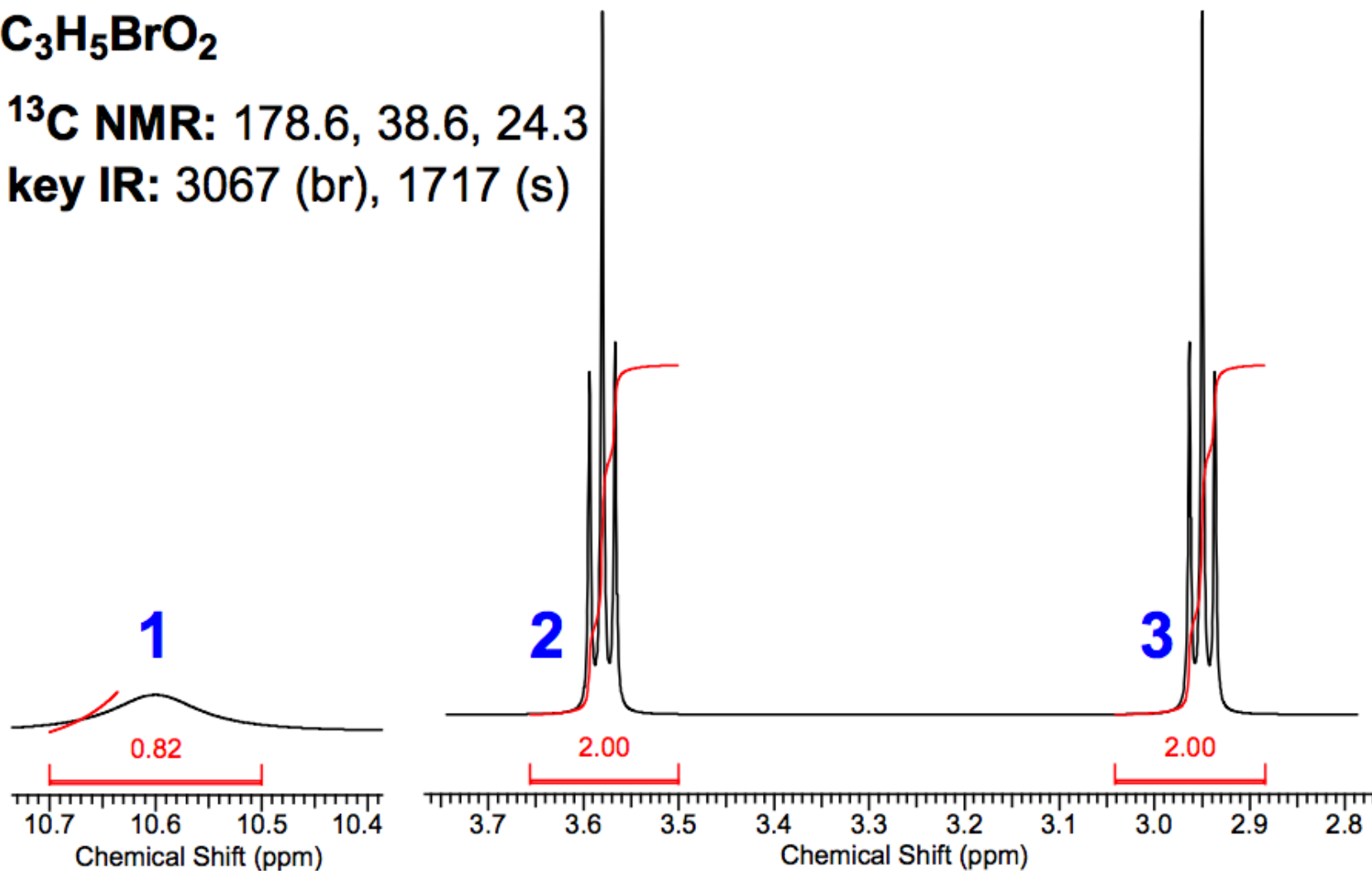


Unknown Compound #1



^{13}C NMR: 178.6, 38.6, 24.3

key IR: 3067 (br), 1717 (s)



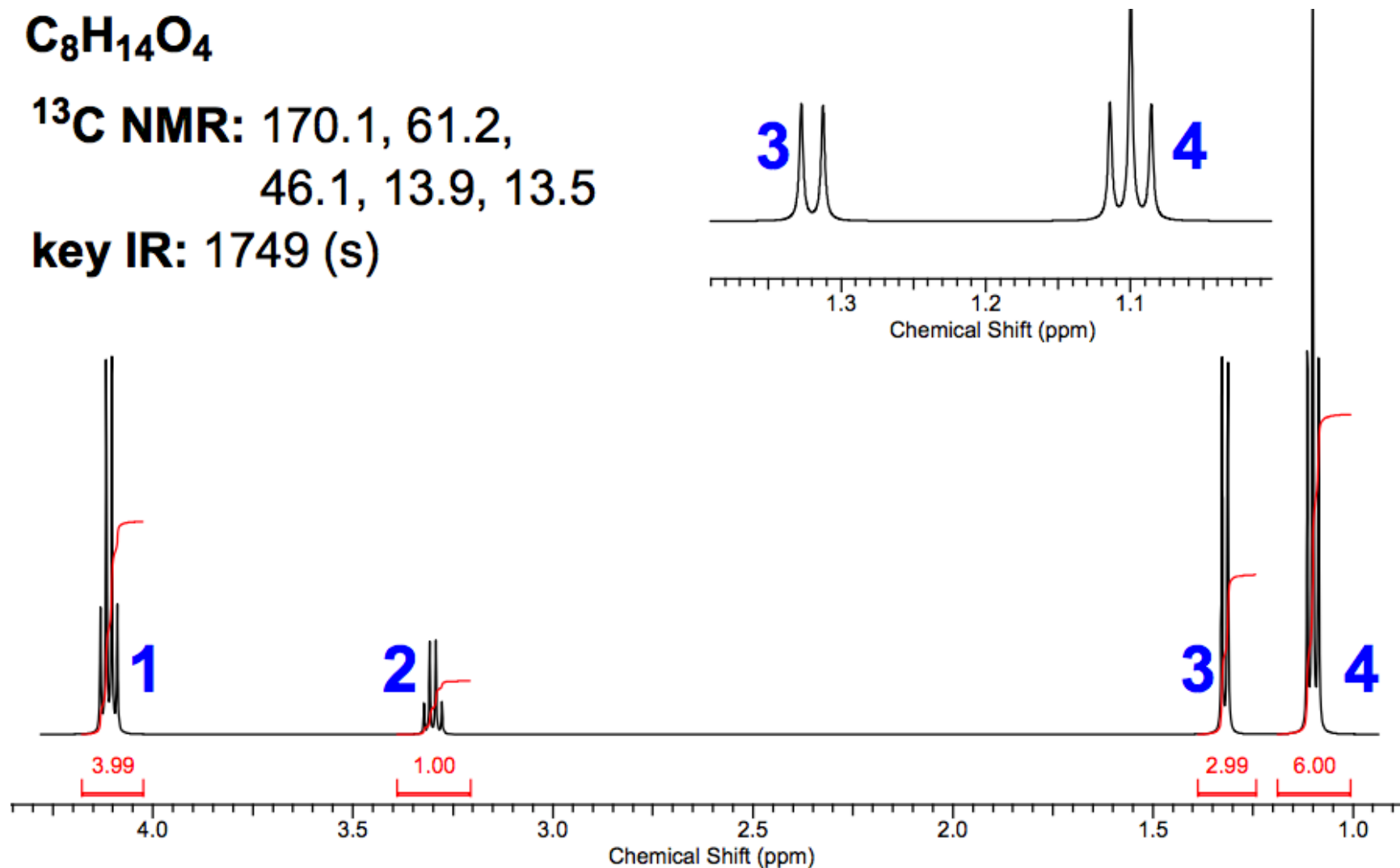


Unknown Compound #2

C₈H₁₄O₄

¹³C NMR: 170.1, 61.2,
46.1, 13.9, 13.5

key IR: 1749 (s)



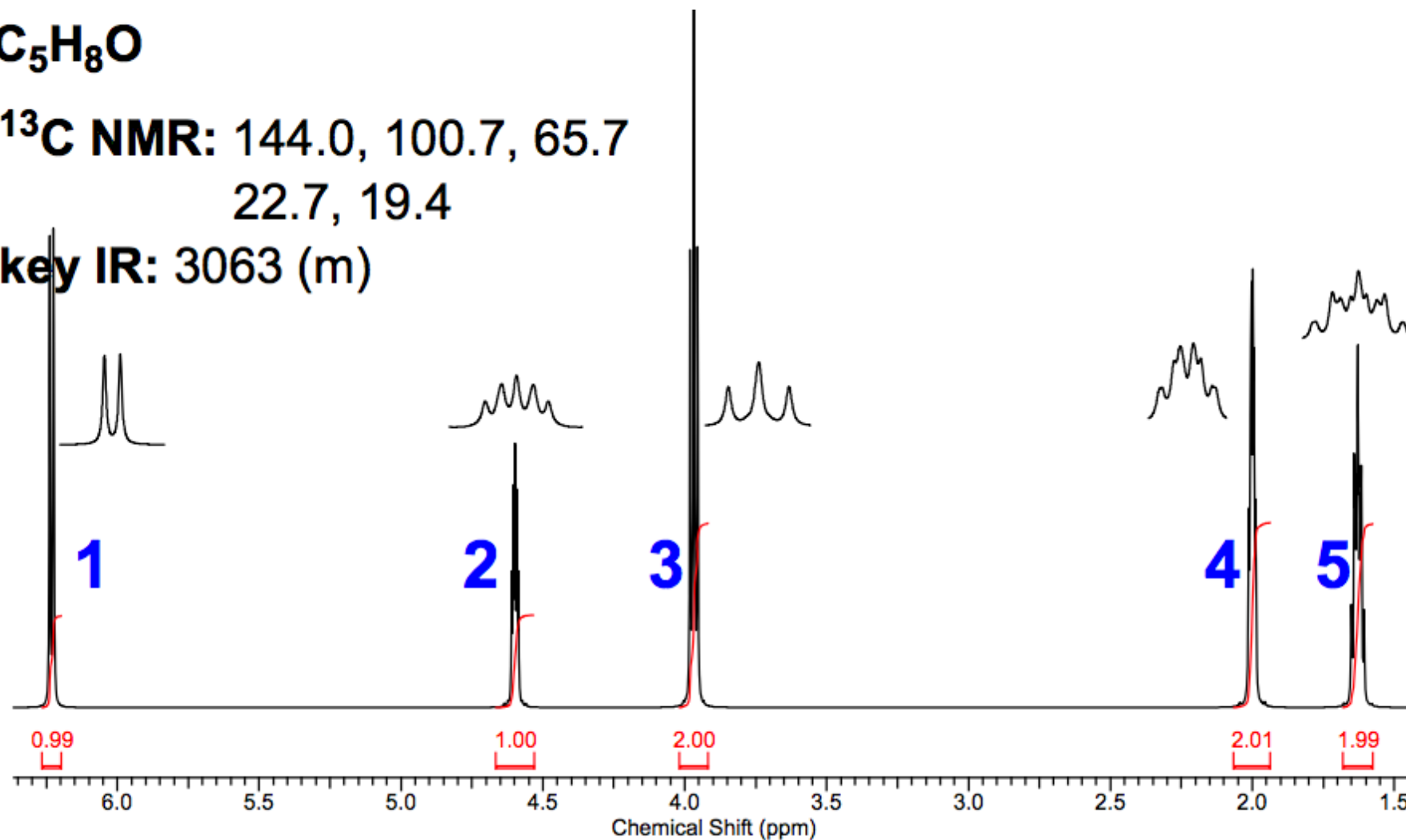


Unknown Compound #3

C₅H₈O

¹³C NMR: 144.0, 100.7, 65.7
22.7, 19.4

key IR: 3063 (m)



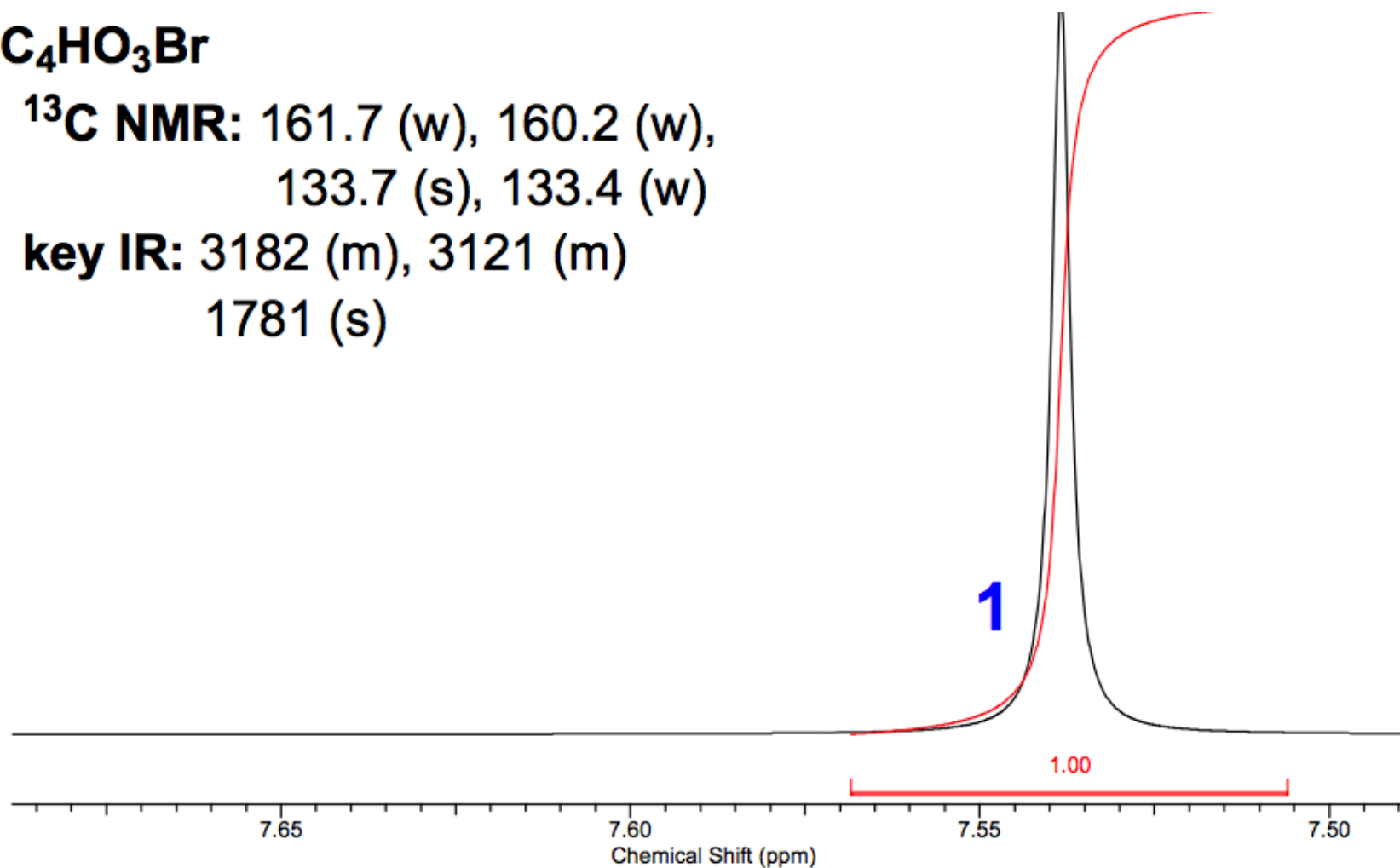


Unknown Compound #4

C₄H₃O₃Br

¹³C NMR: 161.7 (w), 160.2 (w),
133.7 (s), 133.4 (w)

key IR: 3182 (m), 3121 (m)
1781 (s)





Unknown Compound #5

C₅H₇NO₃

¹³C NMR: 20.7 (m), 44.8 (m)
72.9 (m), 165.2 (w), 171.1 (w)
key IR: 3266 (br), 1783 (s)

